

$$D = \{(x_i, y_i)\}_{i=1}^N$$

Linear regression

$$y_i \sim \mathcal{N}(w^T x_i, \sigma^2) \quad (y_i = w^T x_i + \epsilon_i \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2))$$

$$\mathcal{L}(\theta) = \frac{n}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} (y - Xw)^T (y - Xw)$$

$$\nabla \mathcal{L} = 0 \Rightarrow W_{MLE} = (X^T X)^{-1} (X^T y)$$

Consider a feature map or design matrix Φ with rows $\phi(x_n)^T$

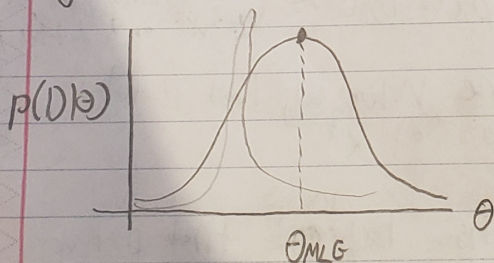
Our new model is $y_i = w^T \phi(x_i) + \epsilon_i$ (why this looks
more non linear
things but not really
 $\phi \rightarrow \text{fixed}$ $N/N \rightarrow \Phi = \Phi_0$)

This is still linear in w so

$$\mathcal{L}(\theta) = \frac{n}{2} \log(2\pi\sigma^2) + \frac{1}{2\sigma^2} (y - \Phi w)^T (y - \Phi w)$$

$$W_{MLE} = (\Phi^T \Phi)^{-1} (\Phi^T y)$$

MLG gives you a single point estimate that maximizes your likelihood



- It cannot say how much the data constrained it (sharpness of the peak)

- It cannot propagate uncertainty into predictions

- It cannot compare models

- A degree 10 poly. can do anything
a degree 3 poly. can more parameters
means it easier to fit to noise (bad)

The alternate is not a better point estimate, its to stop producing a point estimate

Treat w as a random variable

- put a dist. on it before seeing data
- condition on the data

Learning is conditioning not optimizing

Linear
Gaussian

There no objective to minimize and no learning rate

Difficult is moving from opt. to integration

- integrals are usually intractable (except in simple case)

$$p(w|D) = \frac{p(D|w)p(w)}{p(D)}, \quad p(D) = \int p(D|w)p(w)dw$$

The posterior predictive: Distribution over new observation not parameters

$$p(y^* | x^*, D) = \int p(y^* | x^*, w) p(w|D) dw$$

→ Distribution of atomic
from the model y^*
conditional (or given) a new
input x^* and dataset D

prediction: $p(y^* | x^*, w)$ for one value w , this is what the model predicts for an output at x^*

Posterior plausibility: $p(w|D)$ how believable that particular w is after seeing the data

Resd as an average ($\int dw$) of predictions weighted by posterior plausibility

Gaussian family closed
under conditioning
and marginalization

Lecture 2.

$$1) \text{ If } \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \sim \mathcal{N} \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \right)$$

then $p(z_1) = \mathcal{N}(\mu_1, \Sigma_{11})$, $p(z_1|z_2) = \mathcal{N}(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(z_2 - \mu_2),$
(marginal and conditional are still Gaussian) $\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}$)

2)

$$\text{Let } p(z) = \mathcal{N}(z | \mu_z, \Sigma_z) \text{ and } p(y|z) = \mathcal{N}(y | Az + b, \Sigma_y)$$

$$\text{then } p(z|y) = \mathcal{N}(z | \mu_{\text{post}}, \Sigma_{\text{post}}), \quad \Sigma_{\text{post}}^{-1} = \Sigma_z^{-1} + A^T \Sigma_y^{-1} A$$

$$\mu_{\text{post}} = \Sigma_{\text{post}} [A^T \Sigma_y^{-1} (y - b) + \Sigma_z^{-1} \mu_z]$$

$$\text{the marginal is then } p(y) = \mathcal{N}(y | A\mu_z + b, \Sigma_y + A \Sigma_z A^T)$$

Gaussian closed under conditioning and marginalization
which Bayesian inference exist for linear Gaussian system

The posterior for Bayesian Linear regression

$$p(y | x, w) = \mathcal{N}(y | \Phi w, \sigma^2 I), \quad p(w) = \mathcal{N}(w | m_0, S_0) \text{ (model)}$$

Assume σ^2 known, typically $m_0 = 0$, $S_0 = \tau^2 I$

Ignore any terms not involving w

$$\log p(w | D) = -\frac{1}{2\sigma^2} (y - \Phi w)^T (y - \Phi w) - \frac{1}{2} (w - m_0)^T S_0^{-1} (w - m_0) + \text{const}$$

Thy is just $p(y | x, w) p(w)$
 $p(D | w)$ written
more explicitly

Expand

$$-\frac{1}{2\sigma^2} \left[y^T y - 2w^T \Phi^T y + w^T \Phi^T \Phi w \right] - \frac{1}{2} \left[w^T S_0^{-1} w - 2w^T S_0^{-1} m_0 + m_0^T S_0^{-1} m_0 \right]$$

$$= -\frac{1}{2} w^T \underbrace{\left[S_0^{-1} + \sigma^{-2} \Phi^T \Phi \right]}_{\text{quadratic}} w + \underbrace{w^T \left[S_0^{-1} m_0 + \sigma^{-2} \Phi^T y \right]}_{\text{linear}} + \text{const.}$$

Compare with log density of generic Gaussian $N(w | m_N, S_N)$

$$-\frac{1}{2} (w - m_N)^T S_N^{-1} (w - m_N) + \text{const.} = -\frac{1}{2} w^T S_N^{-1} w + w^T S_N^{-1} m_N + \text{const.}$$

Matching coefficients

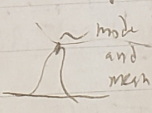
$$S_N^{-1} = S_0^{-1} + \sigma^{-2} \Phi^T \Phi, \quad m_N = S_N (S_0^{-1} m_0 + \sigma^{-2} \Phi^T y)$$

S^{-1} (inverse covariance) is the precision (Messung confidence / high precision $\rightarrow$ low variance)

posterior precision = prior precision + data precision

m_N is a precision weighted average of the prior mean and the data. w/ $m_0 = 0, S_0 = \tau^2 I$

$$S_N = (\tau^{-2} I + \sigma^{-2} \Phi^T \Phi)^{-1}, \quad m_N = \sigma^{-2} S_N \Phi^T y = \left(\Phi^T \Phi + \frac{\tau^2}{\sigma^2} I \right)^{-1} \Phi^T y$$

Gaussian mean = its mode (mode and mean)  ridge estimate w/ $\lambda = \sigma^2 / \tau^2$

The posterior ^{mean} is the MAP estimate

MAP est. = ridge regression = Bayesian post. mean

The new piece is S_N

variance small
precision goes to zero
"im-precise" nothing about w (zero confidence)

Flat prior

$$\tau \rightarrow \infty \quad S_0^{-1} \rightarrow 0 \quad \text{and} \quad S_N \rightarrow \sigma^2 (\Phi^T \Phi)^{-1}$$

$$m_N \rightarrow (\Phi^T \Phi)^{-1} \Phi^T y = \text{WMLE} \quad (\text{recoines MLE})$$

-MLE is MAP w/ the prior

With lots of data: $N \rightarrow \infty$

$$\Phi^T \Phi = \sum_n \phi_n \phi_n^T \approx N E[\phi \phi^T] \quad \left(\begin{array}{l} \tau^2 \text{ become negligible} \\ \Phi^T \Phi \text{ dominates } \sim \text{small data points} \end{array} \right)$$

$$S_N \approx \frac{\sigma^2}{N} (E[\phi \phi^T])^{-1} = O(1/N) \quad \left(\begin{array}{l} \text{more data} \\ \text{smaller } S_N \end{array} \right) \quad \left(\begin{array}{l} -\ln N \\ \text{and grows w/ data} \end{array} \right)$$

The prior contribution
growing $\sigma^{-2} \Phi^T \Phi$ bites at S_0^{-1} is swamped by the

The prior matters when data is scarce and washes out when it's plentiful

Posterior predictive and two kinds of uncertainty

$$\text{We want } p(y^* | x^*, D) = \int N(y^* | w^T \phi^*, \sigma^2) N(w | m_N, S_N) dw$$

Under the posterior $y^* = \underbrace{w^T \phi^*}_{\text{linear in } w} + \underbrace{\varepsilon}_{N(0, \sigma^2)}$, independent
 $w \sim N(m_N, S_N)$

$$p(y^* | x^*, D) = N\left(y^* \mid \underbrace{m_N^T \phi^*}_{\text{opt. weight}}, \underbrace{\sigma^2 + \phi^{*T} S_N \phi^*}_{\text{aleatoric}}\right)$$

epistemic

Algebraic uncertainty (σ^2): Irreducible scatter the model attributes to the observation process

- does not depend on x^* does not shrink as N grows
- collecting data will never reduce it
- It's a model assumption ($\epsilon \sim N(0, \sigma^2)$)

Epistemic uncertainty ($\phi^{*T} \Sigma_N \phi^*$): Uncertainty about w pushed through the model to the prediction

- depends strongly on x^*
- Shrinks like $O(1/N)$ (reducible) by collecting the right pieces of data

One deterministic statement about the world and its fixed. The other is about our own knowledge and is a function of what we have looked at

Epistemic variance answers: given a fixed basis ϕ how much does the data constrain w , it says nothing about whether the basis was right

If you have a basis centred around your training inputs (RBF) then far outside the data/training regime $\phi(x^*) \rightarrow 0$ so $m_N^T \phi^* \rightarrow 0$, $\phi^{*T} \Sigma_N \phi^* \rightarrow 0$ the returns "the answer is zero and I'm certain" (worst outcome)

The posterior here was compatible b/c the prior and likelihood were Gaussian and our model was linear

Instead: Approximate the posterior (VI) or change the model (GPs)

S_N was the big atom of this

~~Allows for variable confidence based on new input~~

S_N lets you see variable confidence based on new input which is used to compute the epistemic uncertainty $\phi^T S_N \phi^*$

The mean is the same MAP as before

A full bellied distribution over the weights instead of a single guess $p(w|D) = \mathcal{N}(w|m_N, S_N)$

In MAP we found $\text{argmax}_w p(w|D)$ (its peak) w/o normalizing it or deriving the full distribution

Now we can sample w